

1 Concept of Data Mining and Its Importance

Data mining is the process of discovering hidden patterns, relationships, trends, and useful knowledge from large datasets using techniques from statistics, machine learning, and databases.

It transforms **raw data** → **meaningful information** → **knowledge for decision making**.

Importance in Modern Organizations

- ✓ Enables data-driven decisions instead of guesswork
- ✓ Helps understand customer behavior and preferences
- ✓ Detects fraud, risks, and unusual activities
- ✓ Improves marketing strategies and business planning
- ✓ Saves time by automatic analysis of large data
- ✓ Supports prediction of future trends

Example: E-commerce companies recommend products based on purchase history.

👉 Data mining helps organizations improve efficiency, profit, and competitiveness.

2 Knowledge Discovery in Databases (KDD) Process

KDD is the complete process of extracting useful knowledge from data.

Data mining is one important step within KDD.

Steps of KDD

1 Data Selection

Relevant data is selected from databases.

2 Data Cleaning (Preprocessing)

Removes noise, missing values, and errors.

3 Data Integration

Combines data from multiple sources into one dataset.

4 Data Transformation

Converts data into suitable format (normalization, aggregation).

5 Data Mining

Applies algorithms to discover patterns and relationships.

6 Pattern Evaluation

Identifies useful and meaningful patterns.

7 Knowledge Presentation

Results are presented using graphs, reports, or visualization.

👉 Output of KDD = actionable knowledge.

3 Types of Data Mining Tasks (with Examples)

Data mining tasks are divided into **predictive** and **descriptive** tasks.

♦ Predictive Tasks (Forecast Future)

✓ Classification

Assigns data into predefined classes.

Example: Loan approval → approved / rejected

✓ Regression

Predicts continuous numeric values.

Example: Predict house price

✓ Time Series Analysis

Analyzes data over time.

Example: Sales forecasting

♦ Descriptive Tasks (Find Hidden Patterns)

✓ Clustering

Groups similar data without predefined labels.

Example: Customer segmentation

✓ Association Rule Mining

Finds relationships between items.

Example: Market basket analysis (Bread → Butter)

✓ Summarization

Provides compact description of data.

Example: Average sales report

Data Mining Functionalities

Data mining functionalities describe the types of knowledge that can be discovered.

✓ **Concept Description**

Summarizes general characteristics of data (characterization, discrimination)

✓ **Frequent Pattern Mining**

Finds commonly occurring patterns in data

✓ **Association Analysis**

Discovers relationships between attributes

✓ **Classification and Prediction**

Builds models to predict future outcomes

✓ **Clustering**

Groups similar objects automatically

✓ **Outlier Analysis**

Detects abnormal or rare data

✓ **Evolution Analysis**

Analyzes trends and changes over time

5 Classification of Data Mining Systems

Data mining systems are classified based on different criteria:

✓ **Based on Data Source**

Relational databases, data warehouses, web data, multimedia data

✓ **Based on Data Model**

Relational model, object-oriented model, hierarchical model

✓ **Based on Knowledge Discovered**

Classification systems, clustering systems, association rule systems, prediction systems

✓ **Based on Techniques Used**

Machine learning, neural networks, statistical methods, artificial intelligence

✓ Based on Application Domain

Finance, healthcare, marketing, web mining

6 Major Issues in Data Mining

1. Data Quality Problems

Real-world data is often incomplete, noisy, inconsistent, or duplicated. Poor-quality data leads to inaccurate results.

2. Data Integration and Heterogeneity

Data comes from multiple sources (databases, files, web) with different formats. Integrating them into a unified dataset is complex.

3. Scalability and Efficiency

Modern organizations store huge volumes of data. Mining large datasets requires efficient algorithms and high computational power.

4. High Dimensionality

Datasets may contain many attributes. This increases processing complexity and may reduce model performance.

5. Privacy and Security

Sensitive data (financial, medical, personal) must be protected. Unauthorized mining may violate privacy laws.

6. Pattern Evaluation

Data mining generates many patterns. Identifying useful, meaningful, and actionable patterns is difficult.

7. Dynamic and Evolving Data

Data continuously changes. Models must be updated regularly to remain accurate.

Conclusion: These issues influence the accuracy, efficiency, and ethical use of data mining systems.

7 Applications of Data Mining in Various Domains

1. Business and Marketing

Customer segmentation, product recommendation, market basket analysis, sales prediction.

2. Banking and Finance

Fraud detection, credit scoring, risk analysis, investment prediction.

3. Healthcare

Disease prediction, patient diagnosis support, treatment effectiveness analysis.

4. Education

Student performance analysis, dropout prediction, personalized learning support.

5. E-commerce

Recommendation systems, customer behavior analysis, demand forecasting.

6. Manufacturing

Quality control, fault detection, production optimization.

7. Telecommunications

Customer churn prediction, network performance optimization.

Conclusion: Data mining improves decision making and operational efficiency across industries.

8 Applying the KDD Process to a Real-World Business Problem

Problem: Improve Sales in a Retail Store

Step 1 — Data Selection

Collect sales transactions, customer records, and product information.

Step 2 — Data Cleaning

Remove duplicates, correct errors, and handle missing values.

Step 3 — Data Integration

Combine data from billing system, inventory system, and customer database.

Step 4 — Data Transformation

Convert raw data into useful formats (monthly sales, purchase frequency).

Step 5 — Data Mining

Apply association rule mining to find frequently purchased product combinations.

Step 6 — Pattern Evaluation

Select meaningful rules (e.g., customers who buy item A also buy item B).

Step 7 — Knowledge Presentation

Present results through reports or visual dashboards.

Result: Better product placement, targeted promotions, and increased sales.

Applying Data Mining Functionalities to a Retail Business Problem

Problem: Increase Profit and Customer Satisfaction

1. Association Analysis

Identify products frequently bought together → cross-selling and bundling.

2. Classification

Classify customers as loyal, regular, or occasional → targeted marketing.

3. Clustering

Group customers based on purchasing behavior → personalized offers.

4. Prediction

Forecast future product demand → improved inventory management.

5. Outlier Detection

Detect abnormal transactions → fraud prevention.

6. Trend and Evolution Analysis

Identify seasonal demand patterns → better planning.

Result: Optimized marketing strategies, efficient inventory, and higher profitability.

10 Data Warehouse Process in Detail

Definition

A **data warehouse** is a centralized repository that stores integrated, historical data to support analysis and decision making.

Characteristics

- Subject-oriented
- Integrated
- Time-variant
- Non-volatile

Data Warehouse Process

1. Data Extraction

Data is collected from multiple operational sources (databases, applications, files).

2. Data Transformation

Data is cleaned, standardized, and converted into a consistent format.

3. Data Loading

Processed data is stored in the data warehouse.

→ These three steps together are called **ETL (Extract–Transform–Load)**.

4. Data Storage and Management

Data is organized for efficient querying and analysis.

5. Data Analysis and Reporting

Users access data using tools such as reporting systems, OLAP analysis, and data mining tools.

Conclusion: A data warehouse provides integrated, historical data that supports strategic decision making.

1 1 OLAP Operations with Multidimensional Data Cube

Definition

OLAP (Online Analytical Processing) allows fast analysis of data stored in a **multidimensional data cube** (dimensions like Time, Product, Location; measures like Sales).

Common OLAP Operations

1. Roll-up (Aggregation)

Summarizes data by climbing up a hierarchy.

Example: City → State → Country sales totals.

2. Drill-down (Detail View)

Opposite of roll-up; shows more detailed data.

Example: Year → Quarter → Month sales.

3. Slice

Selects a single dimension value to form a sub-cube.

Example: Sales for Year = 2025.

4. Dice

Selects a sub-cube using multiple conditions.

Example: Sales for (Product = Laptop, Region = North, Year = 2025).

5. Pivot (Rotate)

Reorients the cube for a different view.

Example: Swap rows and columns in a report.

1 2 Steps of Data Mining Process of Knowledge Discovery (KDD)

Write these steps and draw a flow diagram:

Data Selection → Data Cleaning → Data Integration →

Data Transformation → Data Mining → Pattern Evaluation → Knowledge Presentation

Diagram (write in exam):

Data Sources

↓

Data Selection

↓

Data Cleaning

↓

Data Integration



Data Transformation



Data Mining



Pattern Evaluation



Knowledge/Decision

1 3 Multi-Tiered Architecture of Data Warehouse

Three-Tier Architecture

1. Bottom Tier — Data Warehouse Server

- Data from multiple sources
- ETL process (Extract–Transform–Load)
- Stores integrated historical data

2. Middle Tier — OLAP Server

- Provides multidimensional analysis
- Supports fast query processing

3. Top Tier — Front-End Tools

- Reporting tools
- Dashboards
- Data mining tools
- User interface for decision makers

Diagram (write in exam):

Users/Tools



OLAP Server



Data Warehouse + ETL + Data Sources

Types of Predictive Data Mining

Predictive mining forecasts unknown or future values.

1. Classification

Assigns data to predefined classes.

Example: Email → Spam / Not Spam

2. Regression

Predicts continuous numeric values.

Example: Predict sales amount.

3. Time Series Analysis

Analyzes patterns over time.

Example: Stock price prediction.

4. Prediction Modeling

General forecasting using historical data.

Example: Customer churn prediction.

Types of Descriptive Data Mining

Descriptive mining finds patterns that describe data.

1. Clustering

Groups similar data objects.

Example: Customer segmentation.

2. Association Rule Mining

Finds relationships among items.

Example: Market basket analysis.

3. Summarization

Provides compact data description.

Example: Average monthly sales.

4. Trend and Evolution Analysis

Identifies changes over time.

Example: Seasonal demand patterns.

1 6 Data Mining Task Primitives

These specify **what kind of mining task is to be performed**.

1. Task-Relevant Data

Database, attributes, and data to be mined.

2. Type of Knowledge to be Mined

Classification, clustering, association, prediction.

3. Background Knowledge

Concept hierarchies or domain knowledge.

4. Interestingness Measures

Criteria to evaluate useful patterns (support, confidence).

5. Presentation of Results

Visualization, rules, tables, charts.

1 7 Association Rule Mining (Concept + Example)

Definition

Association rule mining discovers relationships between items in large datasets.

Rule Form

$X \rightarrow Y$

If X occurs, Y also occurs.

Example

In a supermarket dataset:

Customers who buy **bread** often buy **butter**.

Rule: Bread → Butter

Use: Product placement, recommendation systems.

1 8 Support, Confidence, and Lift

Consider rule: Bread → Butter

1. Support

Frequency of transactions containing both items.

Support = (Transactions containing X and Y) / (Total transactions)

2. Confidence

Probability that Y is purchased when X is purchased.

Confidence = (Transactions containing X and Y) / (Transactions containing X)

3. Lift

Measures strength of rule compared to random chance.

Lift = Confidence / Support(Y)

Interpretation:

- Lift > 1 → positive association
 - Lift = 1 → independent
 - Lift < 1 → negative association
-

1 9 Apriori Principle and Its Importance

Apriori Principle

“If an itemset is frequent, then all of its subsets must also be frequent.”

Or

“If an itemset is infrequent, its supersets cannot be frequent.”

Importance

- ✓ Reduces search space
 - ✓ Eliminates unnecessary candidate itemsets
 - ✓ Improves efficiency of frequent itemset mining
 - ✓ Foundation of Apriori algorithm
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2 0 Steps of Apriori Algorithm

Step 1: Generate Candidate 1-itemsets

Count support of each item.

Step 2: Find Frequent 1-itemsets

Remove items below minimum support.

Step 3: Generate Candidate k-itemsets

Join frequent (k-1)-itemsets.

Step 4: Prune Infrequent Itemsets

Remove candidates whose subsets are not frequent (Apriori principle).

Step 5: Repeat

Continue until no new frequent itemsets are found.

Step 6: Generate Association Rules

Form rules from frequent itemsets using confidence threshold.

2 1 FP-Growth Algorithm and Its Working

Definition

FP-Growth (Frequent Pattern Growth) is an efficient algorithm for mining frequent itemsets **without generating candidate itemsets**. It uses a compact structure called an **FP-Tree**.

Working Mechanism

Step 1 — Scan Database

Count support of each item and keep only frequent items ($\geq$ minimum support). Sort them by descending support.

Step 2 — Construct FP-Tree

- Read each transaction
- Insert frequent items in sorted order into a prefix tree
- Share common prefixes; update node counts
- Maintain a **header table** linking same items

Step 3 — Mine the FP-Tree

For each item (from least frequent):

- Build **conditional pattern base** (paths ending with that item)
- Construct **conditional FP-tree**
- Recursively extract frequent itemsets

Advantages

- ✓ No candidate generation
- ✓ Faster than Apriori for large datasets
- ✓ Compresses database effectively

2 2 Comparison: Apriori vs FP-Growth

Feature	Apriori	FP-Growth
Candidate generation	Yes	No
Database scans	Multiple scans	Only two main scans
Data structure	Frequent itemset lists	FP-Tree (compressed)
Performance	Slower for large data	Faster and scalable
Memory usage	Higher (many candidates)	Lower (tree compression)
Complexity	Simple concept	More complex implementation

Conclusion: FP-Growth is generally more efficient, especially on dense and large datasets.

2 3 Market Basket Analysis and Applications

Definition

Market Basket Analysis discovers associations among items purchased together by customers using transaction data.

Example

Customers who buy **bread** often buy **butter** → helps in product placement and bundling.

Applications

- ✓ Product recommendation systems
- ✓ Store layout optimization
- ✓ Cross-selling and promotions
- ✓ Inventory planning
- ✓ E-commerce personalization

Benefit: Improves sales and customer experience by understanding buying patterns.

2 4 Types of Association Rules

1. Multilevel Association Rules

Rules at different levels of abstraction using concept hierarchies.

Example:

Level 1: Dairy → Bakery

Level 2: Milk → Bread

Use: Analyze patterns from general to specific levels.

2. Multidimensional Association Rules

Rules involving multiple attributes (dimensions).

Example:

Age = 20–30 $\wedge$ Income = High → Buys Laptop

Use: Combines demographic and behavioral attributes.

3. Single-Dimensional Rules

Only one attribute (usually items).

Example: Bread → Butter

4. Boolean vs Quantitative Rules

- **Boolean:** presence/absence of items
 - **Quantitative:** considers numeric values (e.g., quantity, price range)
-

2 5 Applying Apriori to Generate Frequent Itemsets

Example Transaction Database

TID Items

T1 Bread, Milk

T2 Bread, Diaper, Beer, Eggs

T3 Milk, Diaper, Beer, Cola

T4 Bread, Milk, Diaper, Beer

T5 Bread, Milk, Diaper, Cola

Assume **minimum support = 2**.

Step 1 — Frequent 1-Itemsets (L1)

Count supports:

Bread = 4

Milk = 4

Diaper = 4

Beer = 3

Cola = 2

Eggs = 1 **✗** remove

L1 = {Bread, Milk, Diaper, Beer, Cola}

Step 2 — Candidate 2-Itemsets (C2) → Frequent 2-Itemsets (L2)

Count supports:

{Bread, Milk} = 3

{Bread, Diaper} = 3

{Milk, Diaper} = 3

{Diaper, Beer} = 3

{Milk, Beer} = 2

{Diaper, Cola} = 2

(others < 2 removed)

L2 = all pairs above

Step 3 — Candidate 3-Itemsets (C3) → Frequent 3-Itemsets (L3)

{Bread, Milk, Diaper} = 2

{Milk, Diaper, Beer} = 2

{Bread, Diaper, Beer} = 2

L3 = these 3-itemsets

Step 4 — Stop

No larger frequent itemsets possible.

Final Frequent Itemsets:

L1, L2, and L3 above → used to generate association rules.

2 6 Construction of an FP-Tree (Example)

Sample Transaction Dataset

T1: {Bread, Milk}

T2: {Bread, Diaper, Beer, Eggs}

T3: {Milk, Diaper, Beer, Coke}

T4: {Bread, Milk, Diaper, Beer}

T5: {Bread, Milk, Diaper, Coke}

Assume **min support = 2**

Step 1 — Count Item Supports

Bread=4, Milk=4, Diaper=4, Beer=3, Coke=2, Eggs=1 ✗ remove
Order by support: **Bread, Milk, Diaper, Beer, Coke**

Step 2 — Reorder Transactions (keep frequent items)

T1: Bread, Milk

T2: Bread, Diaper, Beer

T3: Milk, Diaper, Beer, Coke

T4: Bread, Milk, Diaper, Beer

T5: Bread, Milk, Diaper, Coke

Step 3 — Build FP-Tree (prefix sharing)

(Null)

/ \

Bread:4 Milk:1

|

Milk:3

|

Diaper:3

/ \

Beer:2 Coke:1

(Header table links all nodes of the same item.)

Idea: Common prefixes are merged; counts are stored in nodes.

2 7 Determining Strong Association Rules (Support & Confidence)

Formulas

$\text{Support}(X \rightarrow Y) = \text{support}(X \cup Y)$

$\text{Confidence}(X \rightarrow Y) = \text{support}(X \cup Y) / \text{support}(X)$

Example

From 5 transactions:

$$\text{support}(\{\text{Milk}\}) = 4/5 = 0.8$$

$$\text{support}(\{\text{Milk, Diaper}\}) = 3/5 = 0.6$$

Rule: Milk $\rightarrow$ Diaper

$$\text{Confidence} = 0.6 / 0.8 = \mathbf{0.75 (75\%)}$$

If min support ≥ 0.5 and min confidence $\geq 0.7 \rightarrow$ **Strong rule.**

2 8 Rule Generation from Frequent Itemsets

Example Frequent Itemset

{Bread, Milk, Diaper} with support = 2/5

Generate Rules

1. Bread, Milk $\rightarrow$ Diaper
2. Bread, Diaper $\rightarrow$ Milk
3. Milk, Diaper $\rightarrow$ Bread
4. Bread $\rightarrow$ Milk, Diaper (and so on)

Evaluate One Rule

Milk, Diaper $\rightarrow$ Bread

$$\begin{aligned} \text{Confidence} &= \text{support}(\{\text{Bread, Milk, Diaper}\}) / \text{support}(\{\text{Milk, Diaper}\}) \\ &= (2/5) / (3/5) = \mathbf{2/3 \approx 0.67} \end{aligned}$$

Keep rules meeting min confidence.

2 9 Association Rule Mining in Real-World Business

Retail Store

- Identify items bought together $\rightarrow$ bundling, shelf placement.

E-commerce

- Personalized recommendations (“Customers who bought X also bought Y”).

Banking

- Detect unusual combinations → fraud detection.

Healthcare

- Discover co-occurring symptoms or treatments.

Telecom

- Service usage combinations → targeted plans.

Outcome: Better marketing, inventory planning, and customer satisfaction.

3 0 Support for {Milk, Diaper}

Dataset (5 transactions):

T1 {Milk, Bread}

T2 {Milk, Diaper, Beer, Eggs}

T3 {Milk, Diaper, Beer, Coke}

T4 {Bread, Milk, Diaper, Beer}

T5 {Bread, Milk, Diaper, Coke}

Transactions containing {Milk, Diaper} → T2, T3, T4, T5 = 4

Support({Milk, Diaper}) = $4 / 5 = 0.8$ (80%)

3 1 Support and Confidence Calculation

Support({Milk, Diaper})

= $4 / 5 = 0.8$ (80%)

Confidence for Rule Milk → Diaper

support({Milk}) = appears in T1, T2, T3, T4, T5 = 5

support({Milk, Diaper}) = 4

Confidence = $4 / 5 = 0.8$ (80%)

3 2 Confidence for Rule Bread → Milk

Given:

support(Milk) = 30%

$\text{support}(\text{Bread}) = 60\%$

$\text{support}(\text{Milk}, \text{Bread}) = 40\%$

$\text{Confidence}(\text{Bread} \rightarrow \text{Milk})$

$= \text{support}(\text{Milk}, \text{Bread}) / \text{support}(\text{Bread})$

$= 40\% / 60\%$

$= \mathbf{0.67 (\approx 66.7\%)}$

3 3 Support, Confidence, and Lift for $A \rightarrow B$

Given:

Total transactions = 100

A appears = 40

B appears = 30

A and B together = 20

Support

$\text{Support}(A \rightarrow B) = \text{support}(A \cap B)$

$= 20 / 100 = \mathbf{0.20 (20\%)}$

Confidence

$\text{Confidence}(A \rightarrow B) = \text{support}(A \cap B) / \text{support}(A)$

$= 20 / 40 = \mathbf{0.50 (50\%)}$

Lift

$\text{Lift}(A \rightarrow B) = \text{Confidence}(A \rightarrow B) / \text{support}(B)$

$\text{support}(B) = 30 / 100 = 0.30$

$\text{Lift} = 0.50 / 0.30 = \mathbf{1.67}$

Interpretation: $\text{Lift} > 1 \rightarrow$ positive association.

3 4 Apriori with Minimum Support = 20%

Transactions:

$T1 = \{A, B, C\}$

$T2 = \{A, C\}$

$T3 = \{A, B\}$

$T4 = \{B, C\}$

$T5=\{A,B,C\}$

Min support = 20% of 5 = **1 transaction**

Step 1 — Frequent 1-Itemsets (L1)

$A=4, B=4, C=4 \rightarrow$ all frequent

Step 2 — Frequent 2-Itemsets (L2)

$AB=3, AC=3, BC=3 \rightarrow$ all frequent

Step 3 — Frequent 3-Itemsets (L3)

$ABC=2 \rightarrow$ frequent

Final Frequent Itemsets:

$L1=\{A,B,C\}$

$L2=\{AB, AC, BC\}$

$L3=\{ABC\}$

3 5 Frequent Pattern Mining

Definition

Frequent pattern mining discovers **itemsets, sequences, or patterns** that appear frequently in a dataset above a minimum support threshold.

Types

- Frequent itemsets
- Sequential patterns
- Association rules

Applications

Market basket analysis, recommendations, trend discovery.

Goal: Identify regularities in data to support decisions.

3 6 FP-Growth Algorithm (with Neat Diagram)

Idea

Mines frequent itemsets using an **FP-Tree** without candidate generation.

Working Steps

1. Scan database and find frequent items.
2. Build FP-Tree (compressed representation).
3. Mine tree recursively using conditional pattern bases.

Diagram (write in exam)

Transactions → Count Items → Build FP-Tree →

Conditional Pattern Base → Conditional FP-Tree →

Frequent Patterns

Key Point: Faster than Apriori due to no candidate generation.

3 7 FP-Tree: Structure and Components

Definition

An **FP-Tree (Frequent Pattern Tree)** is a compact prefix-tree structure that stores frequent items and their counts.

Components

1. **Root Node** → Null node
2. **Item Nodes** → store item name + count
3. **Paths** → represent transactions
4. **Header Table** → links identical items
5. **Node Links** → connect same-item nodes

Purpose: Compress database and enable efficient mining.

3 8 Steps to Construct an FP-Tree

1. Scan database and compute item supports.
2. Remove infrequent items (below min support).
3. Sort frequent items by descending support.

4. Read each transaction:
 - keep frequent items only
 - insert into tree following sorted order
 - update counts
5. Maintain header table with node links.

Result: Compact tree representing all frequent items.

Advantages of FP-Growth over Apriori

- ✓ No candidate itemset generation
- ✓ Fewer database scans
- ✓ Faster for large datasets
- ✓ Better memory efficiency (tree compression)
- ✓ Handles dense data effectively
- ✓ Reduces computational cost

Conclusion: FP-Growth is more scalable and efficient.

Mining Frequent Patterns from an FP-Tree

Process

Step 1 — Start from Least Frequent Item

Select item from header table.

Step 2 — Find Conditional Pattern Base

Collect all prefix paths leading to that item.

Step 3 — Construct Conditional FP-Tree

Build a smaller tree using pattern base.

Step 4 — Extract Frequent Patterns

Combine item with patterns found recursively.

Simple Illustration

FP-Tree

↓

Select Item

↓

Conditional Pattern Base

↓

Conditional FP-Tree

↓

Frequent Itemsets

Key Idea: Divide-and-conquer mining without generating candidates.

1 Applying FP-Growth to Find Frequent Itemsets (Example)

Sample Database (min support = 2)

T1: {A, B, D}

T2: {B, C, E}

T3: {A, B, C, E}

T4: {B, E}

Step 1 — Count Item Supports

A=2, B=4, C=2, D=1 , E=3

Frequent items: **B, E, A, C** (sorted by support → B > E > A > C)

Step 2 — Reorder Transactions

T1: B, A

T2: B, E, C

T3: B, E, A, C

T4: B, E

Step 3 — Build FP-Tree

(Null)

|

B:4

/ \

A:1 E:3

/ \

C:1 A:1

\

C:1

Step 4 — Mine Frequent Patterns

Frequent itemsets (support ≥ 2):

{B}, {E}, {A}, {C}

{B,E}, {B,A}, {B,C}, {E,C}

{B,E,C}

4 2 Conditional FP-Tree Construction (Illustration)

Using previous FP-Tree, find patterns for item **C**.

Step 1 — Find Paths Ending with C

Paths leading to C:

- B $\rightarrow$ E $\rightarrow$ C
- B $\rightarrow$ E $\rightarrow$ A $\rightarrow$ C

Step 2 — Conditional Pattern Base for C

{B, E} : 1

{B, E, A} : 1

Step 3 — Build Conditional FP-Tree

(Null)

|

B:2

|

E:2

Step 4 — Frequent Patterns with C

{C, B}, {C, E}, {C, B, E}

3 Pattern Growth Technique

Pattern growth mines frequent itemsets **by expanding patterns recursively** from the FP-Tree.

Steps

1. Build FP-Tree from database.
2. Select an item from header table.
3. Extract conditional pattern base.
4. Build conditional FP-Tree.
5. Extend pattern with new items.
6. Repeat recursively.

Key Idea

Instead of generating candidates (Apriori), patterns grow directly from compressed tree structure.

Result: Efficient mining of frequent itemsets.

Construct FP-Tree and Find Frequent Itemsets

Given Transactions (min support = 2)

T1: {A, B}
T2: {B, C, D}
T3: {A, C, D, E}
T4: {A, D, E}
T5: {A, B, C}

Step 1 — Count Supports

A=4, B=3, C=3, D=3, E=2
All are frequent.

Step 2 — Sort Items by Support

$A > B = C = D > E$

Step 3 — Reorder Transactions

T1: A, B

T2: B, C, D

T3: A, C, D, E

T4: A, D, E

T5: A, B, C

Step 4 — Construct FP-Tree

(Null)

/ \

A:4 B:1

/ \

B:2 D:1

| \

C:1 E:1

|

C:1

\

C:1

(Counts reflect shared prefixes; structure may be drawn cleanly in exam.)

Step 5 — Frequent Itemsets (support ≥ 2)

Single items:

{A}, {B}, {C}, {D}, {E}

Pairs:

{A,B}, {A,C}, {A,D}, {A,E}, {B,C}, {C,D}, {D,E}

Triples:

{A,D,E}

5 Construct FP-Tree (Min Support = 30%)

Given Dataset (5 transactions)

T1 = {Milk, Bread}

T2 = {Bread, Diaper, Beer, Eggs}

T3 = {Milk, Diaper, Beer, Coke}

T4 = {Bread, Milk, Diaper, Beer}

T5 = {Bread, Milk, Diaper, Coke}

Minimum support = 30% of 5 = **2 transactions**

Step 1 — Count Item Supports

Bread = 4

Milk = 4

Diaper = 4

Beer = 3

Coke = 2

Eggs = 1  remove (infrequent)

Frequent items sorted by support:

Bread, Milk, Diaper, Beer, Coke

Step 2 — Reorder Transactions (remove Eggs)

T1: Bread, Milk

T2: Bread, Diaper, Beer

T3: Milk, Diaper, Beer, Coke

T4: Bread, Milk, Diaper, Beer

T5: Bread, Milk, Diaper, Coke

Step 3 — Construct FP-Tree

(Null)

/ \

Bread:4 Milk:1

|

Milk:3

|

Diaper:3

/ \

Beer:2 Coke:1

(Header table connects same-item nodes.)

This tree represents all frequent items compactly.

Frequent Patterns using FP-Growth (Min Support = 2)

Dataset

T1={A,B,C}

T2={A,C}

T3={B,C}

T4={A,B,C}

T5={A,B}

Step 1 — Count Supports

A=4, B=4, C=4 → all frequent

Step 2 — Frequent Itemsets

Single Items

{A}, {B}, {C}

2-Itemsets

AB = 3

AC = 3

BC = 3

3-Itemsets

ABC = 2

Final Frequent Patterns

{A}, {B}, {C}

{A,B}, {A,C}, {B,C}

{A,B,C}

(All satisfy minimum support ≥ 2)

Class/Concept Description

(Data Characterization & Data Discrimination)

Definition

Class/Concept description summarizes the general features of a target class of data. It helps understand **what kind of data belongs to a class** and how it differs from others.

1. Data Characterization

Meaning

Summarizes the **general characteristics of a target class**.

Purpose

To describe common properties of data belonging to one class.

Example

Describe features of high-value customers:

- High income
- Frequent purchases
- Premium product preference

Output Forms

✓ Tables

✓ Rules

- ✓ Charts
 - ✓ Summaries
-

2. Data Discrimination

Meaning

Compares a **target class with contrasting classes** to find differences.

Purpose

To identify what makes one class different from another.

Example

Compare:

High-value customers vs Low-value customers

Differences may include:

- Purchase frequency
 - Spending amount
 - Product category
-

Key Difference

Data Characterization	Data Discrimination
Describes one class	Compares multiple classes
Finds common features	Finds distinguishing features
Summary of target data	Comparison with contrasting data

Conclusion

Class/concept description helps in **understanding data behavior**, identifying patterns, and supporting decision making through summarization and comparison.